

Customer Shopping Behavior Analysis

3,900 transactions · 18 features · Python · SQL · Power BI



Dataset at a Glance

3,900

Transactions

18

Columns

37

Missing Values

Review Rating only

Customer Data

Age, Gender, Location, Subscription Status

Purchase Details

Item, Category, Amount, Season, Size, Color

Data Preparation in Python

01

Loading & Data Exploration

02

Data Cleaning

Median imputation by product category

03

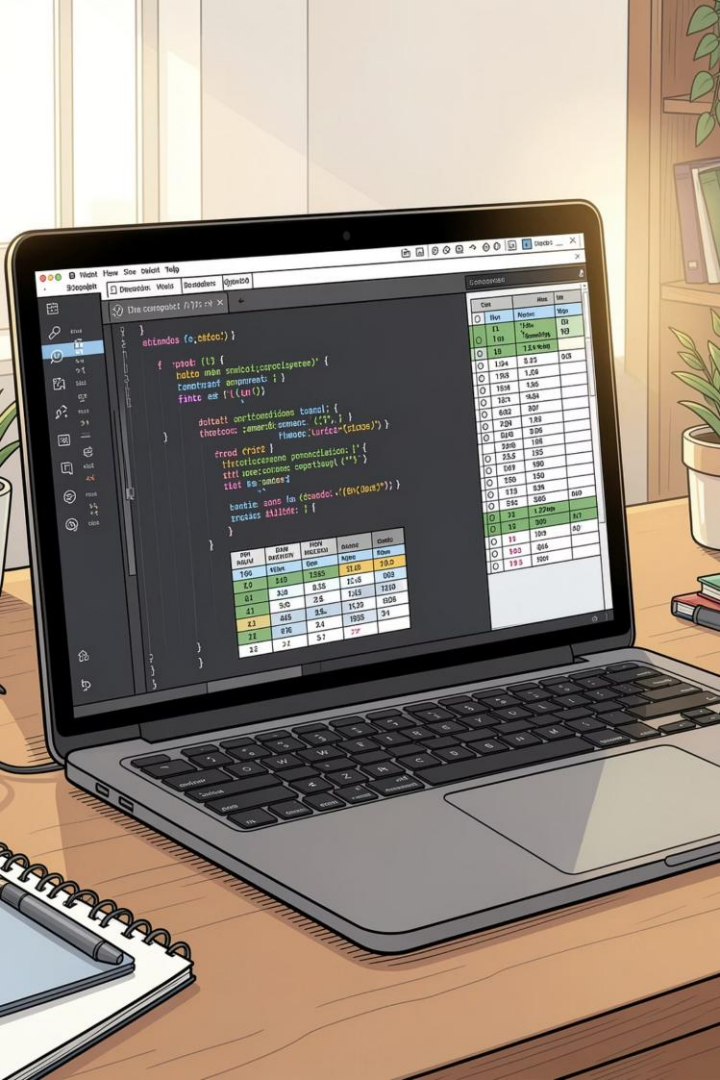
Data Transformation

Group Age Ranges · New column addition · Column name standardization

04

Load to PostgreSQL

Data Analysis



Revenue by Gender

	gender text 	revenue numeric 
1	Female	75191
2	Male	157890

Key Question

Compared total revenue generated by male vs. female customers.

Spending & Discounts

High-Spending Discount Users

Customers using discounts who still spent **above average**.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88
Total rows: 839		Query complete 00:00:0

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

Discount-Dependent Products

Top 5 products with highest **discounted purchase rate**.

Product Insights

Top 5 Products by Rating

Highest average review ratings across all categories.

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessories	Jewelry	171
2	2	Accessories	Sunglasses	161
3	3	Accessories	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

Top 3 Products per Category

Most purchased items within each product category.

Shipping & Subscriptions

	shipping_type text 🔒	round numeric 🔒
1	Standard	58.46
2	Express	60.48

Shipping Type Comparison

Average spend: **Standard vs. Express** shipping.

	subscription_status text 🔒	total_customers bigint 🔒	avg_spend numeric 🔒	total_revenue numeric 🔒
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

Subscribers vs. Non-Subscribers

Average spend and total revenue by subscription status.

Customer Segmentation & Loyalty

Segments

Classified as **New**, **Returning**, or **Loyal** based on purchase history.

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

Repeat Buyers & Subscriptions

Customers with **>5 purchases** more likely to subscribe.

Revenue by Age Group

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

Age Group Contribution

Total revenue calculated per age segment to identify highest-value demographics.

Insights

Customers & Spending 3,900 customers | Avg. purchase: \$59.76 | Avg. rating: 3.75/5

Subscriptions 73% are non-subscribers — major growth opportunity

Revenue by Category Clothing leads, followed by Accessories, Footwear, and Outerwear

Age Group Young Adults generate the most revenue and purchases; Adults rank last in sales

Key Takeaway Converting non-subscribers and targeting Young Adults could significantly boost revenue



Business Suggestions

Convert Non-Subscribers — 73% unsubscribed; offer discounts or free trials to drive sign-ups

Loyalty Programs
Reward repeat buyers

Double Down on Clothing — highest revenue and sales category; prioritize stock and promotions

Re-engage Adults — lowest in sales; use targeted promotions to activate this segment

Target Young Adults — top segment in revenue and purchases; prioritize in campaigns and loyalty programs

Targeted Marketing
Focus on high-revenue age groups